ANTO JOEIS

AI Engineer · LLM Agents, RAG & Automation · Local LLMs on GPU · Computer Vision

New Taipei City, Taiwan — **relocating to Singapore** (Employment Pass required) | +886-979-927-914 | antojoeis@gmail.com | linkedin.com/in/anto-joeis

PROFESSIONAL SUMMARY

AI engineer with 5+ years turning operational problems into secure, production AI applications and automation. Builds LLM agents that call internal APIs as tools, RAG over internal documentation, and an AI-assisted vulnerability-management pipeline (SBOM in CI, LLM triage of CRITICAL/HIGH CVEs). Deploys and serves local models on GPU infrastructure, and ships with Docker, CI/CD and hardened APIs, taking each solution from prototype to handover.

CORE SKILLS

- **AI applications:** LLM agents & tool calling, RAG & embeddings, prompt engineering, local model deployment (Ollama, Qwen, LLaMA), GPU inference
- **Security:** Vulnerability management: SBOM (Syft), Dependency-Track, CVE triage & remediation; API hardening, SSH automation (paramiko)
- **Engineering:** Python, Flask, REST APIs, Swagger, JavaScript/Node.js, PostgreSQL, MongoDB
- **DevOps:** Docker, CI/CD (GitLab CI), systemd, automated provisioning (PXE, dnsmasq)
- **ML:** TensorFlow, OpenCV, Darknet/YOLO, OpenVINO, CUDA
- **Languages:** English (professional), Mandarin (conversational), Tamil (native)

PROFESSIONAL EXPERIENCE

R&D Software Engineer — IEI Integration Corp

May 2021 – Present

New Taipei City, Taiwan · Telecom Product R&D

- Built an **AI-assisted supply-chain security pipeline**: automated SBOM generation (Syft in GitLab CI) plus an LLM agent (Claude CLI) that triages CRITICAL/HIGH CVEs from Dependency-Track and publishes security reviews to an internal portal.
- Built an **LLM agent that calls internal APIs as tools**: chatbot → local LLM → REST API → serial console → device under test, on a single Flask backend. Test engineers now run **15+ hardware-test workflows** across multiple product lines in plain language.
- Added a **RAG pipeline** (embeddings-based retrieval) that grounds the test chatbot's LLM answers in internal test documentation.
- Achieved **~100× faster LLM inference** than CPU-only by deploying Qwen and LLaMA via Ollama with NVIDIA GPU acceleration on an on-premise edge server.
- **Improved deployment reliability by 75%** by consolidating multiple OS provisioning pipelines into one dnsmasq DHCP + TFTP server for simultaneous multi-device boot, and hardening the Flask boot-config API.
- **Reduced data latency by 40% and downtime by 15%** with containerised on-premise data infrastructure (Docker, MongoDB replication); led REST API integration across projects.
- **Cut manufacturing defects by 30%** by fine-tuning pretrained vision models on a custom defect dataset (TensorFlow/Darknet) and serving them in a camera capture → inference → machine control REST pipeline backed by self-hosted MongoDB.
- Integrated backend REST APIs to control networked video-orchestration devices (matrix, video-wall, multi-viewer; up to 32 sources) and ran cross-platform end-to-end hardware testing.

Automation Testing Engineer — Top Union Electronics Corp

Nov 2020 – Apr 2021

Taiwan · Functional Testing & QA

- Implemented functional testing procedures, raising operational efficiency by 20% and reducing product defects by 30%.
- Performed failure analysis and component-level repair, cutting average issue-resolution time by 30%; authored SOPs adopted team-wide.

Networking Maintenance Engineer — Globe Computers

Oct 2014 – Nov 2015

India · System & Network Maintenance

- Delivered networking and desktop support; produced error-trend reports that enabled proactive maintenance.

EDUCATION

M.S., Electrical Engineering & Communication — National Tsing Hua University, Taiwan (2016 – 2020)
Thesis: Smart Public Transport Using Deep Learning — custom thermal-image dataset (IR sensors, Raspberry Pi), transfer learning, real-time seat-occupancy detection deployed on edge microcomputers.

B.E., Electronics & Communication — Anna University, Chennai, India (2010 – 2014)
Research: hierarchical remote-sensing image analysis with segmentation and SVM classification.

CERTIFICATIONS

Generative AI with LLMs — Google Cloud / Coursera (2025) · Python — Udemy (2022)